Dean Wu
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Education

- **University of Toronto** — Mississauga, Canada
  Bachelor of Science in Computer Science, Statistics and Mathematics; — *Sep. 2023 – Present*

Experience

- **China Telecom (TeleAI)** — Beijing, China
  LLM Engineer Intern — *May 2026 – Aug. 2026*
  - **WeCom AI Assistant**: Built a LangChain/LangGraph ReAct agent for WeCom (WeChat Work) that creates, reads, edits, and searches company documents and spreadsheets using LLMs (DeepSeek and Kimi), streaming replies over a persistent WebSocket connection with per-user conversation memory by using vector databases mixed with JSON metadata filtering.
  - **Agent Skills & Context Design**: Built a modular skill architecture (docs, sheets, search, memory, file uploads) on a shared JSON tool dispatcher, and used Google's OKF method to give the LLM short Markdown skill summaries, so it knew its tools without being overloaded by lengthy descriptions, greatly speeding up responses.
  - **Agent Gym**: Contributed the harness-side agent loop for an internal agent gym (sandboxed RL environment) used for agent training and data synthesis: the agent plans a JSON TODO list of subtasks with shell check commands, the harness drives each step, then runs a final verification pass and re-prompts failed items with stdout/stderr feedback until checks pass.
  - **Coding Harness**: Helped design an internal coding-agent harness, contributing core features such as agent loop and auto-dreaming.
- **UTASR (University of Toronto Autonomous Scale Racing)** — Toronto, Canada
  Software & Optimization Engineer — *Oct 2025 - May 2026*
  - **Nvidia Jetson Orin NX**: Architecting a stable, high-performance runtime on a headless Jetson Orin NX , using Python multiprocessing and IPC queues to manage parallel ML and Data Engineering workloads.
  - **Automation**: Developing a complete system automation and monitoring framework integrating Bash, cron, tegrastats, and htop to ensure low-overhead logging (<1ms) and rapid watchdog restarts (<5s).
  - **Remote System**: Implementing a robust remote orchestration system via SSH , delivering automated scripts for system startup, health checks, and repository updates.
- **Game Making Team | Loyola C.S.S.** — Mississauga, Canada
  Lead Software Engineer/Project Manager — *Dec. 2022 - May 2023*
  - **Project Management**: Managed the project's development using Scrum methods. This involved planning work in focused sprints and leading regular team check-ins to meet all deadlines with a high-quality result
  - **Software Implementations**: Lead the game development by architecting the project in GameMaker Studio 2, coding core systems (such as movement, fighting, quests) in GML, and leading quality assurance in map design and collaborative team code reviews.

Projects

- **Discord AI**: Developed a multimodal Discord AI bot with autonomous agents for web search, stock/crypto tracking, weather forecasting, and gaming statistics using Python, LangChain, Vision AI, and Upstash Redis for persistent memory
- **Project Mura**: Developed a top-down 2D action game using GameMaker Studio 2 with custom GLSL shader programming.
- **deanwu.me**: Developed and deployed a responsive personal portfolio website using React, React Router, and Vite with custom CSS.

Programming Skills

- **Languages**: Python, Java, GameMaker Language, RISC-V Assembly, C++
- **Technologies**: LangChain, LangGraph, ChromaDB, asyncio, Pydantic, pytest, React, Vite, NPM, Upstash, NodeJS, Raspberry Pi, Arduino, Git